A3-PS008-TC065

Network Management Agent for Reviewed Configuration changes.

Mukul Goyal, Ishika Pandey
Faculty mentor - Dr. Tejaswi Khanna

goyalmukul110@gmail.com
ishikapandey.2807@gmail.com
tejaswi.khanna@gmail.com

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Problem Description

Router changes typed by hand reach the live network untested and unreviewed.

- A typo or wrong next-hop takes effect immediately.
- A command valid on one router can conflict with another router's routes, causing a routing loop or a black hole that no single device reports.
- No test step, no mandatory second reviewer, no record of who approved what.
- Recovery after a bad change depends on someone remembering the old configuration.

Background

What is needed to follow the solution

Intent-Based Networking

The operator states the desired outcome; software derives the commands. [1]

Static Routing

Each router forwards to a next-hop for a destination. Correctness depends on all routers working together.

Loop & Blackhole

Loop: next-hops lead back to a router already visited.

Blackhole: a router on the path has no route.

Digital Twin & NETCONF

Test proposed changes in a software model before touching real devices.

NETCONF workflow: Propose → Validate → Commit, or Roll back.

Literature & RFC Review

RFC 9315, Intent-Based Networking, defines intent and its lifecycle: translate, verify, deploy, and assure. Our pipeline follows it.

RFC 6241, NETCONF: candidate datastore, validate, rollback-on-error. We reproduce this workflow on a platform without NETCONF.

RFC 7950, YANG 1.1: typed, constrained configuration schemas. Our intent schema follows the same style.

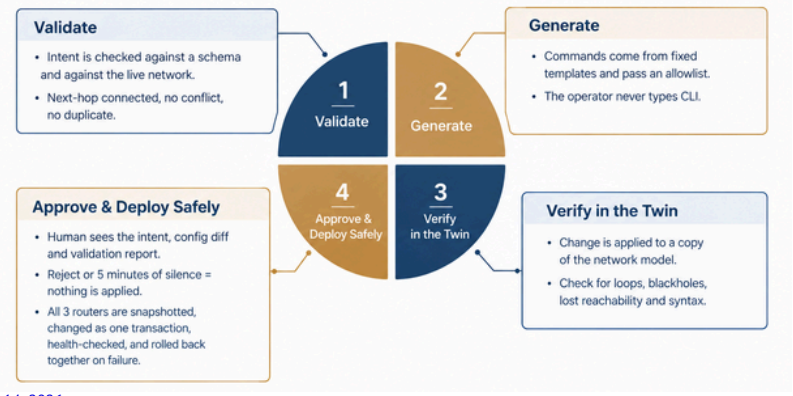
Batfish shows that you can find loops and black holes by analysing configuration offline, before deployment.

Gap: existing tools verify configuration. None combines intent translation, verification, a mandatory human approval gate and automatic rollback in one auditable flow.

Proposed Solution

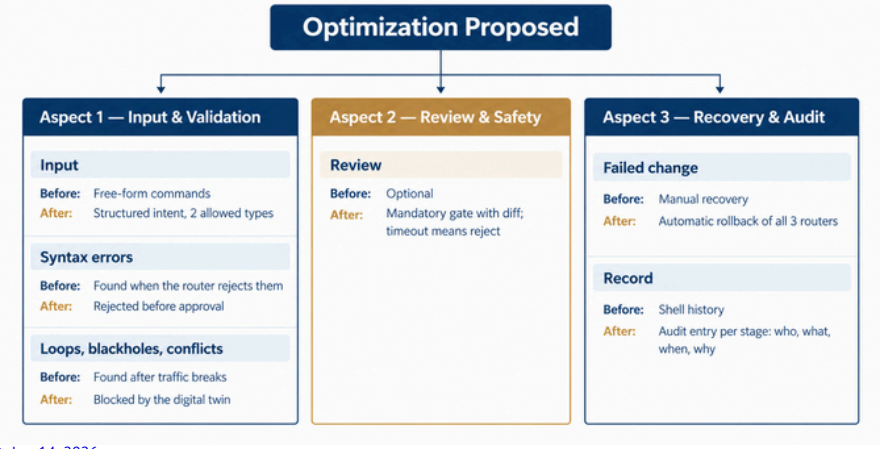
A human-reviewed agent: intent in, verified and approved change out.

Lab: 3 FRRouting routers in a triangle on Containerlab [5][6]. Intents: ADD_ROUTE, REMOVE_ROUTE.



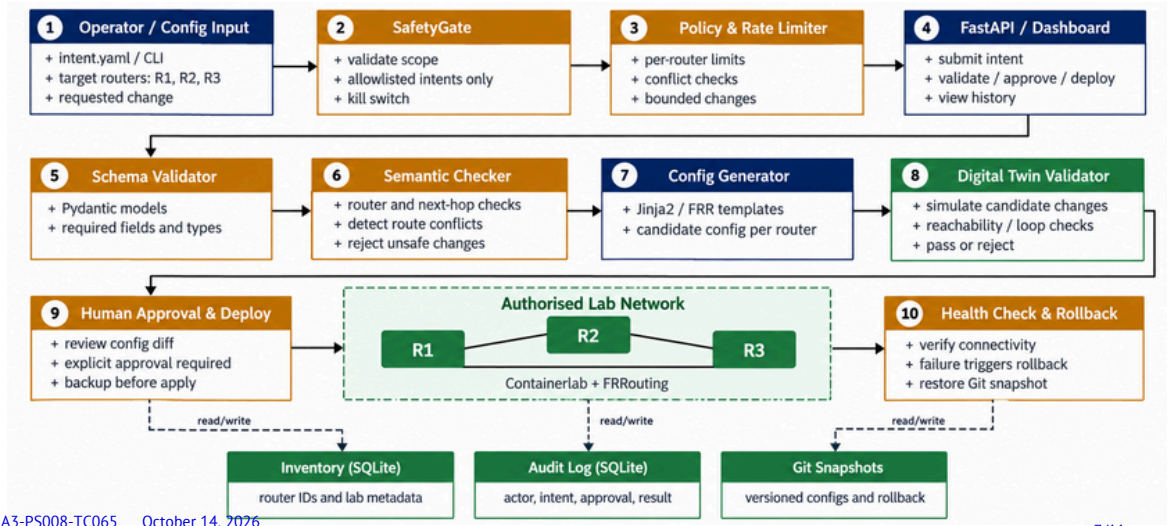
Optimization Proposed

The proposed agent shifts network configuration from reactive, manual operations to a validated, human-approved and recoverable workflow.



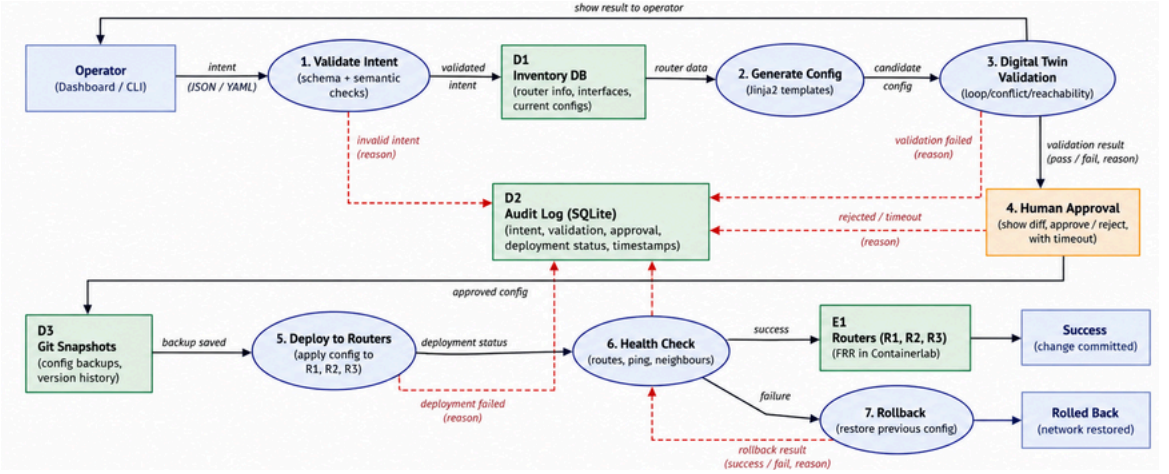
Solution Architecture & Design

Blue : Application Modules • Orange : Safety Gates • Green : Authorised lab and data



Data Flow Diagram

Intent → validate → generate → twin → human approval → snapshot → deploy → health check → SUCCESS or ROLLED BACK. Each failure exits to the audit log without touching the routers.



Deployment Timeline

Method: build the safety core first; each milestone has a pass condition before the next begins.

Days	Milestone	Done when
1–3	3-router lab, routes configured by hand	Every LAN pings every other LAN
3–5	Intent schema and config generator	Valid intent gives the right command; invalid gives a clear reason
5–8	Digital twin	A looping change is blocked and the path is named
8–10	Deploy, health check, rollback	A forced failure restores all 3 routers
10–14	Integration, dashboard, tests	12+ scenarios pass across the 4 PS categories
14–16	Report, demo, submission	Report in repo, pseudocode shared with mentor

Bibliography

[1] A. Clemm, L. Ciavaglia, L. Z. Granville, J. Tantsura. “Intent-Based Networking - Concepts and Definitions.” RFC 9315, October 2022.

[2] R. Enns, M. Bjorklund, J. Schoenwaelder, A. Bierman. “Network Configuration Protocol (NETCONF).” RFC 6241, June 2011.

[3] M. Bjorklund. “The YANG 1.1 Data Modeling Language.” RFC 7950, August 2016.

[4] A. Fogel et al. “A General Approach to Network Configuration Analysis.” USENIX NSDI, 2015.

[5] FRRouting Project. “FRRouting User Guide.” https://docs.frrouting.org

[6] Containerlab. “Containerlab Documentation.” https://containerlab.dev

Thank You for Your Attention!